

USE-CASE FLOW SPECIFICATION

Use-Case Name: Generate Audit Export

System: Centralized Audit Trail Compliance Engine

Primary Actor: Compliance Officer

Supporting Actor: Security Auditor

Description: This use case enables a Compliance Officer to search for required audit records and generate a secure audit export. The export contains masked records and cryptographic integrity hashes so that a Security Auditor can later verify whether the records have been modified.

Trigger: The Compliance Officer selects the "Generate Audit Export" option.

Preconditions:

1. The Compliance Officer must be authenticated.
2. The Compliance Officer must have permission to access and export audit records.
3. Audit records must be available in the system.

Postconditions:

1. An audit export containing the selected records is generated.
2. A SHA-256 integrity hash is included for every exported record.
3. The export operation is recorded in the system's audit trail.

Main Success Scenario:

1. The Compliance Officer selects the "Generate Audit Export" option.
2. The system displays filters such as date range, source application, event type and user.
3. The Compliance Officer enters the required filter values.
4. The system validates the entered filter values.
5. The system calculates a SHA-256 integrity hash for every selected record.
6. The system generates an export containing the selected records and an integrity-hash manifest.
7. The system makes the generated audit export available for download.
8. The system records the Compliance Officer's identity, export time, selected filters and export result in its audit trail.
9. The use case ends successfully.

Alternate Flow – No Matching Audit Records:

5a. The system finds no audit records matching the entered filters.

5a.1. The system displays the message “No matching audit records found.”

5a.2. The system allows the Compliance Officer to modify the search filters.

Exception Flow – Export Generation Failure:

10a. The system is unable to generate the export because of a storage or processing error.

10a.1. The system displays an export-failure message.

10a.2. The system records the failure and its cause in the audit trail.